



02451 Introduction to Machine Learning

Week 1: Introduction

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3 February 2026

DTU Compute, Technical University of Denmark



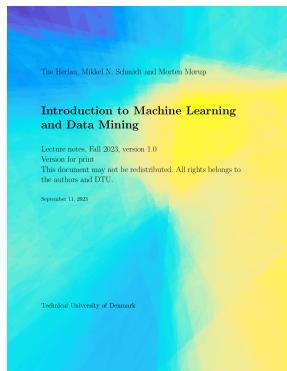
Today

Feedback group of the day:

Marius Andersen, Ana Ariyan Aydin, Jakub Bouzan, Lucas Loua Buhelt, Sai Renuka Chandrasekaran, Matthes Moritz Fogtmann, Katarina Heüman Funch, Victor Valentin Glyrskov, Leyi Han, Marie Nongnapas Holldorff, Victor Kragballe, Sakarias Haugen Kragh, Kristian Kristiansen, Sze Yuen Ku, Samuel Kjær Mackie, Nour Marak, Lorenzo Marcolungo, Sara Li Møller, Andreas Bang Mørch, Sebastian Hvid Olavsgaard, Ida-Marie Colding Olsen, Hannibal Hilden Otte, Lucas Beenfeldt Reüter, Marcus Christian Wad Sackett, Caroline Mathilde Schibelfeldt, Sindri Dagur Sindrason, Sophia Alexandra Suhr Sólstein, Berit Tikenberg, Rosa Skovsted Walter, Noah Paul Sylvester-Hvid Westheimer.

Reading material:

Chapter 1, Chapter 2, Chapter 7



Lecture Schedule

1 Introduction, data, and visualization

3 February: C1, C2, C7

2 Summary statistics, similarity, nearest neighbors and decision trees

10 February: C4, C12, C9

3 Computational linear algebra and PCA

17 February: C3

4 Probability theory, Bayes

and Naive Bayes

24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)

5 Bayesian inference and linear models

3 March: C5, C6, C8

6 Overfitting and cross-validation

10 March: C10

7 Performance evaluation and AUC

17 March: C11, C16 (Assignment 4-6 due on 19 March)

8 Artificial Neural Networks and Optimization

24 March: C15

9 Bias/Variance and ensemble methods

7 April: C14, 17

10 K-means and hierarchical clustering

14 April: C18 (Assignment 7-9 due on 16 April)

11 Mixture models and density estimation

21 April: C19, C20

12 Representation learning

28 April: Material provided separately

13 Recap and discussion of the exam

5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

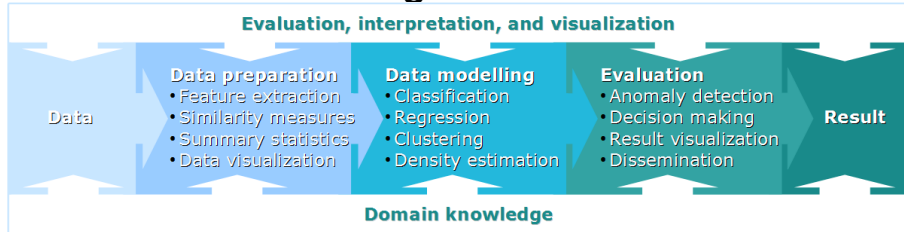
Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Plan for today:

- Lecture 1 (13:00 – ~15:00)
 - **Introduction**
 - What is machine learning?
 - This course and expectations
 - Pre-test
 - **Overview of the course topics**
 - **What is data?**
 - Dataset types (tabular, relational, sequential)
 - Attribute types
 - Data quality and issues - missing, noise, corrupted
 - Data in a vector space
 - **Visualization**
- Exercises in Python (~15:00–17:00)

What is machine learning?



Learning Objectives

- Understand what machine learning is and can be used for
- Understand the types of data, their attributes, transformations of attributes and data issues
- Understand how to represent data modalities, such as text and images, as vectors
- Understand and apply a wide range of data visualization approaches

Alan Turing (1946)

Alan Turing
(1912-1954)



- Universal computing
- Proposed machines should learn like children

We are not in a position to answer if a machine can think because the terms machine and think are undefined. Rather, we should ask if a machine can imitate a human.

(the imitation game)

Arthur Samuel (1959)

Arthur Samuel
(1901-1990)



- Samuels wrote a checkers playing program
 - Program played 10000 games against itself
 - Learned value of each board position by considering the resulting score

Machine learning: (The) field of study that gives computers the ability to learn without being explicitly programmed

Tom Michell (1999)

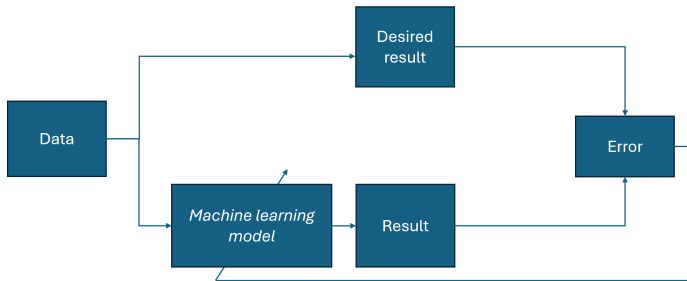
A well-posed learning problem: A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at the tasks in T , as measured by P , improves with experience E

- Checkers example
 - E : Playing 10'000 games
 - T : Playing checkers
 - P : Win or loose

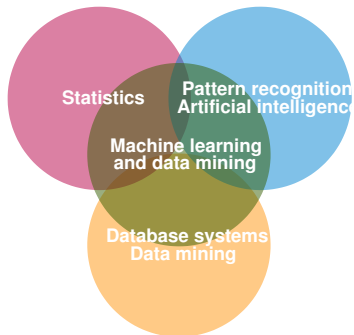


Tom Michell

An engineering view

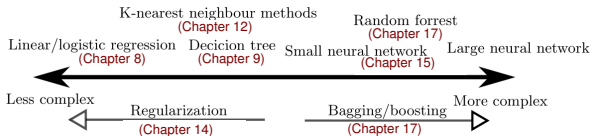
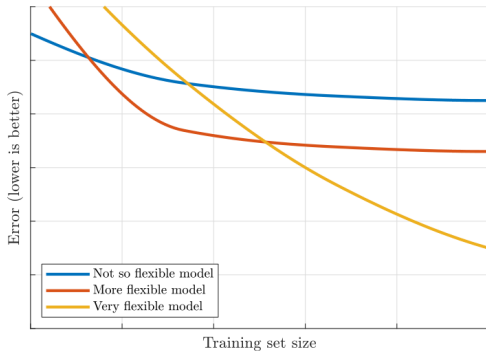


Machine-learning as a separate field



- Focus on a *learning algorithms* (rather than search, pathfinding, etc.)
- De-emphasize explicit knowledge representations, etc.
- Gradual improvements (training time, amount of data)
- *General* algorithms (or algorithmic ideas)

Machine-learning as a research area



Machine learning advances...

- 2022-2026, ChatGPT, Claude, Mistral, Llama, DeepSeek Large Language Models (LLMs)
- 2020, 2024, alphafold Outperforms all state-of-art expert systems for protein structure prediction.
- 2020, Breast cancer Outperforms radiologists in breast-cancer detection (Nature)
- 2019, Lung cancer Outperform six doctors with a 5% reduction in false negatives (DeepMind)
- 2019, Starcraft 1v1: OpenAI deep reinforcement learning exhibit high-level performance in SC2
- 2018, BERT: Superhuman performance on the SQuADv1.1 wikipedia question-answer task
- 2018, alphago: superhuman chess/go learned from scratch
- 2017, Texas hold'em no limit: Libratus (Carnegie Mellon) beats top professional
- 2017, Alpha Go: Superhuman Go by reinforcement learning + imitation of expert games
- 2016, libreading : Superhuman libreading from Oxford and Google Deepmind
- 2016, conversational speech: Microsoft research demonstrate superhuman speech recognition
- 2016, Geoguessing Google PlaNet win 28 of 50 rounds; median localization error of 1132km vs. 2321km
- 2015, closed-world image recognition Microsoft report error of 4.94% on ImageNet vs. 5.1% for top-human labeler
- 2015, Atari RL Google Deepmind obtain better-than-expert human performance on many Atari video games
- 2012, ImageNet AlexNet massively improved performance on closed-world image recognition
(arguably preceded by a 2011 effort to recognise road signs by Schmidhuber et al)

List inspired by: Finn's blog

Why now?

Scientific Advances in algorithmic ideas

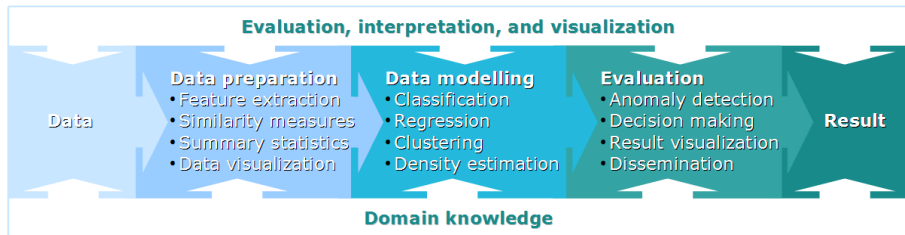
Empirical Increased availability of large/good datasets from *relevant* domains

Technological Faster computers, especially graphics processing units (GPU)

Social Libraries which automate routine tasks; increased sharing of code, etc.

Economical Greatly increased resource allocation and economic incentives

This course (vs other machine learning courses)



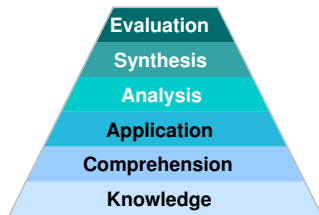
Who should take this course?

Prerequisites

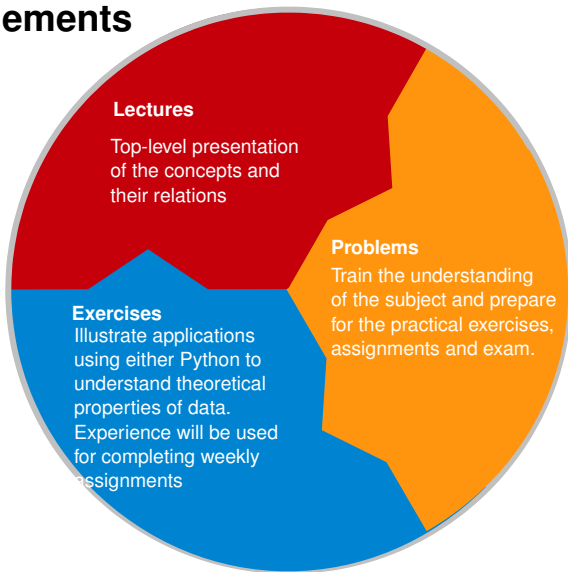
- **Mathematics** (linear algebra, multivariate calculus, differentiability, integration). E.g. 01001/01002/01003/01004/01005
- **Statistics or probability** (axioms of probability, random variables, expectations, marginalization, product rule, Bayes rule, distributions, densities, p-value, paired t-test, confidence intervals). E.g. 02402/02403 (ideally 02405)
- **Introduction to Programming** in Python (including basic knowledge of numerical aspects and tools, e.g., NumPy in Python). E.g. 02002/02101/02102/02525

Learning Objectives

- Explain the major steps involved in data modeling from preparing the data, modeling the data, to evaluating and disseminating the results.
(Knowledge)
- Discuss key machine learning concepts such as feature extraction, cross-validation, generalization and over-fitting, prediction, curse of dimensionality, and the bias-variance trade-off.
(Comprehension)
- Match practical problems to standard data modeling problems such as dimensionality reduction, regression, classification, density estimation and clustering.
(Comprehension and Application)
- Explain how a relevant set of machine learning methods works. Describe assumptions, strengths, and limitations of relevant machine learning methods.
(Comprehension and Analysis)
- Apply, modify, and implement central aspects of machine learning algorithms in Python
(Application and synthesis)
- Apply visualization techniques and statistics to evaluate model performance, identify patterns and data issues.
(Analysis)
- Select, combine and modify data modeling tools in order to analyze data and disseminate the results of the analysis.
(Application, Analysis, Synthesis and Evaluation)



Course Elements



Assessment

The assessment consists of two components

- A four-hour written multiple-choice exam with negative marking (no aids with two sheets of A4 paper with **handwritten** notes). No calculator.
 - **Important:** Take notes throughout the semester!
- Assessed assignments (individual) in weeks 1-12.
 - **Important:** 8 of 12 assignments must be approved to be able to pass the course.

Final grade based on an **overall assessment** of assignments (all 12) and written exam. The written exam is weighted more than the reports ($\approx 90\%$ vs $\approx 10\%$)

Course format

Lecture session [2 hours per week]

- In-person: main auditorium 116-81 [275 seats] and streamed to 116-83 [100 seats].
- Online: live webcast via Zoom (link on DTU Learn). Most exercise rooms can be used during lectures.
- Offline: lectures recorded and uploaded to Panopto (via **Video recordings** on DTU Learn) before the next lecture.

Exercise session [2 hours per week]

- Physically - find rooms & information on DTU Learn (recommended).
- Online - only very limited assistance, contact a TA via Piazza / e-mail (expect some delay).

Detailed structure and activities

- **Workload:** DTU's **nominal** workload for a 5 ECTS course is **9 hours per week** during the 13-week period and 140 hours in total.
- **Structure and study-activities :**
 - Lecture session [2 hours per week]:
 - **High-level lectures** including quizzes and in-class discussions (formative, i.e. not assessed)
 - Supervised **exercises** (formative) [2 hours per week]
 - Focus on the central aspects of the exercises, not the project work.
 - Self-study, preparation & project work [5 hours per week]
 - We assign **readings, homeworks, quizzes** and provide **old exams** to help you study more effectively (not assessed).
 - **Assignments (assessed)**, you apply the taught theory and programming experience to unseen problems.
 - Online help/assistance via Piazza.
 - **Exam (assessed)** [4h + preparation (19h)]

Individual vs group Learning

- We encourage group learning during lectures, exercises, and in part for the assignments.
 - **During lectures** You are encouraged to work together and discuss the online Quiz questions.
 - **During exercises** Each exercise consists of computer exercises relevant to the reported work and a conceptual multiple-choice question from a previous exam.
 - **Individual assignments** You are allowed to discuss high-level aspects of the problem in groups, but the submission must be evidence of an individual effort.

Online help

- Piazza: <https://piazza.com/dtu.dk/spring2026/02451> (Sign up!)
- Use Discussion Forum (i.e. Piazza) for 24/7 help
- Ensures that everyone have access to the same information
 - **Bad: very general questions, i.e., can you explain GMM?**
Good: Here is what I understand, but I don't get equation ...
 - **Bad: error without context, i.e., I tried to do a PCA, but I got an error that the matrix has the wrong size.**
Good: What language are you using and what is the error; code which produced the mistake; what did you try to accomplish?
 - **Bad: Can you explain the PCA question in the Fall 2009 exam?**
Good: Insert a screenshot of fall 2009 exam, explain what part of the solution is unclear

Additional machine learning courses

This course (02451) is a **prerequisite** for:

- (02445 Project work for AI & data – currently only for KID students)
- (02462 Signals and data — currently only for KID students)
- 02456 Deep learning
- 02465 Reinforcement Learning
- 02476 Machine Learning operations
- 02471 Machine learning for signal processing
- (024xx LLM Operations - currently a special course)
- 02477 Bayesian machine learning
- 02460 Advanced Machine Learning (requires 02456, 02471, 02477)
- + several domain-specific/applied machine learning courses at DTU

Pretest

- The purpose is to assess your background in order to adjust the presentation and measure your learning.
- Some of the questions will be hard and may seem unfair. Do not Google it or use generative AI. We want to know, if you know.
- **Not part of your assessment. We do not look at individual results.**
- Start during the break and finish during the exercise session.

Go to: DTU Learn > 02451Spring 2026 > Quizzes > Pretest

14:05

- video

- 8 of 12

Break

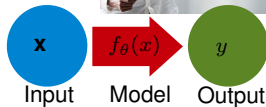
Next up: *An overview of machine learning*

Machine learning tasks

Supervised - predictive tasks

Use some variables to predict unknown or future values of another variable

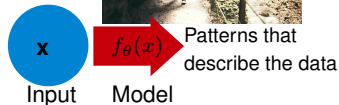
- **Classification:** *Determine which class a data object belongs to based on the input*
 - Discrete output
- **Regression:** *Determine the output value from the input*
 - Continuous output



Unsupervised - descriptive tasks

Find (human-interpretable) patterns that describe the data

- **Clustering**
 - Discover group structure in data
- **Anomaly Detection** (e.g., via density estimation)
 - Find data objects that are abnormal
- **Representation learning**
 - Discover the latent representations of the data



Classification: Definition

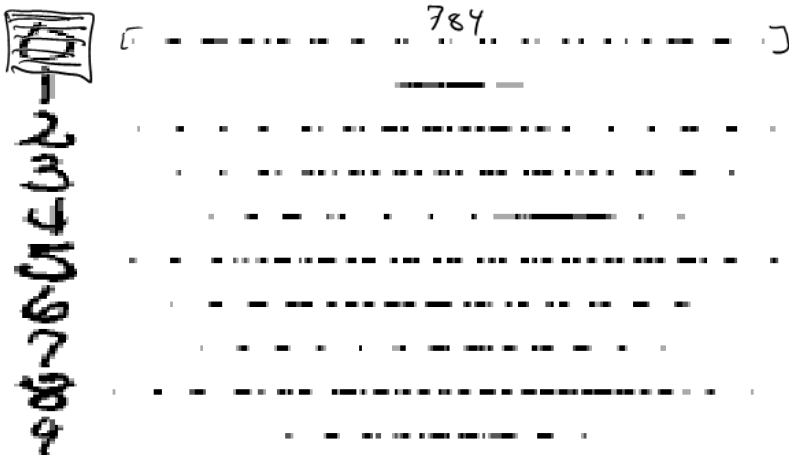
- Given a collection of data objects (**training set**)
 - Each object has associated features (aka variables, attributes)
 - Each object belongs to a certain class
- Define a **model** for the class given the other features
- Goal: Assign a class label to a **previous unseen object**

Example: Image classification

Training set										Classify	
0	1	2	3	4	5	6	7	8	9	5	?
0	1	2	3	4	5	6	7	8	9	2	?
0	1	2	3	4	5	6	7	8	9		
0	1	2	3	4	5	6	7	8	9		
0	1	2	3	4	5	6	7	8	9		

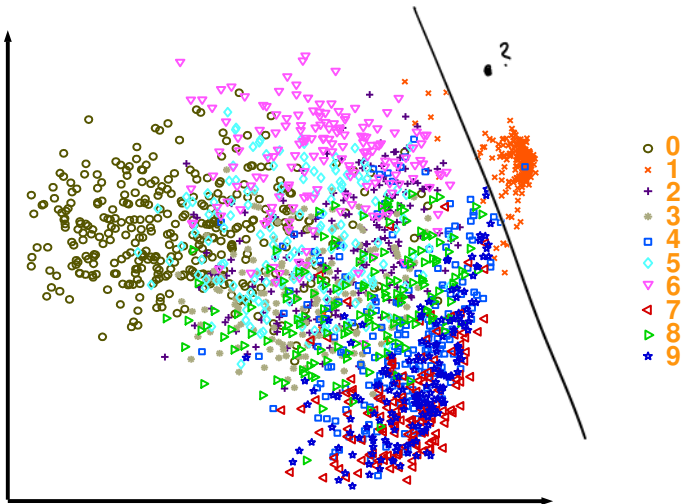
Example: Image classification

Data representation



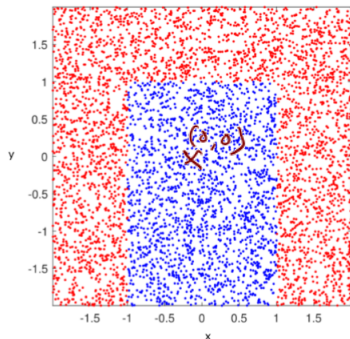
Example: Image classification

Visualization



Quiz 01: Decision rules

$$\max\{-10, 0\} = 0$$



The figure shows an example classification problem consisting of a large number of observations (x, y) along with their class (red and blue).

A *decision rule* is just a function which takes an (x, y) coordinate and outputs either the red or the blue class. Suppose we define:

$$z_1 = \max\{0, x - 1\} + \max\{0, -1 - x\}$$

$$z_2 = \max\{0, -1 + y\}$$

Which of the following *decision rules* solve the problem?

- A. If $z_1 = z_2 = 0$ classify as blue and otherwise as red
- B. If $z_1 = z_2 = 1$ classify as blue and otherwise as red
- C. If $z_1 = 1$ and $z_2 = 0$ classify as blue and otherwise as red
- D. If $z_1 = 0$ and $z_2 = 1$ classify as blue and otherwise as red
- E. Don't know

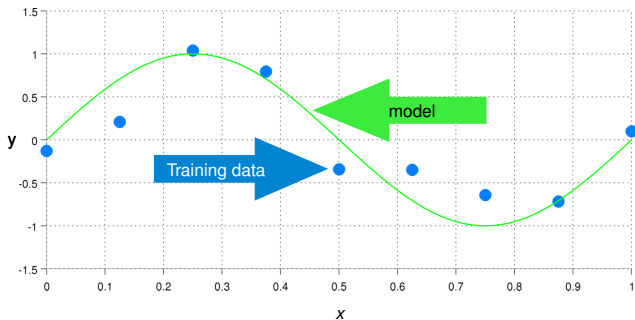
Solution:

The first option will give the correct boundary. To see this, check out $(x, y) = 0$. In this case $z_1 = 0$ and $z_2 = 0$. Therefore, only the first option gives the correct answer.

Actually, this decision rule is a (slightly obscured) example of a neural network with RELU units. We will learn more about neural networks later in this course.

Regression: Definition

- Given a collection of data objects (**training set**)
 - Each object has associated features (aka variables, attributes)
 - Each object is associated with to **countinuous valued variable**
- Define a **model** for the variable given the features
- **Goal**: Predict the value of the variable for a **previously unseen input**



Example: Regression

- Predict **sales amounts** of a new product based on **advertising expenditure**.
- Predict **wind velocity** as a function of **temperature**, **humidity**, and **air pressure**
- Predict **stock market index** based on historical **index values** and **market indicators**.

A few possible functions:

- 1-dimensional input

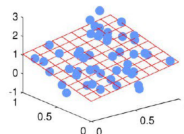
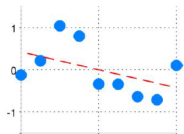
$$f(x; \mathbf{w}) = w_0 + w_1x$$

- 2-dimensional input

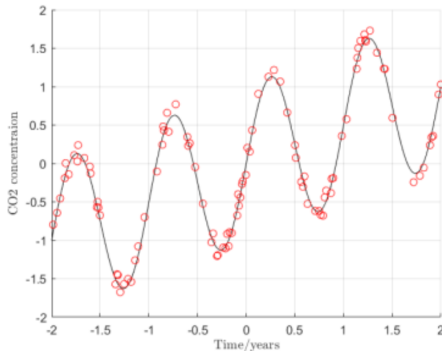
$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1x_1 + w_2x_2$$

- 2-dimensional input, polynomial feature transformation/basis expansion

$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2$$



Quiz 02: Regression



The figure shows an example regression problem where the CO₂ concentration is measured as a function of the time of year.

We wish to come up with a prediction rule $y = f(x)$ where y is the relative CO₂ concentration and x is the time of year. Which of the following functions would be a good candidate?

- A. $y = 0.5x + \cos(x)$
- B. $y = -0.5x + \cos(x)$
- C. $y = 0.5x + \sin(2\pi x)$
- D. $y = -0.5x + \sin(2\pi x)$
- E. Don't know

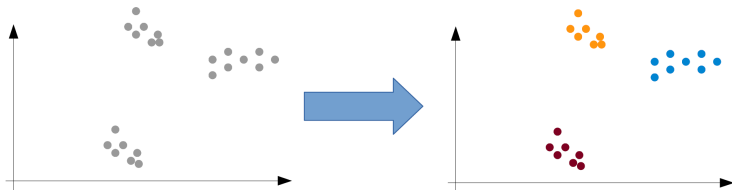
Solution:

The right answer is $y = 0.5x + \sin(2\pi x)$. The functions involving cosines can be ruled out since they have the wrong period; meanwhile the function has an

upwards trend which means it must involve the term $0.5x$.

Clustering: Definition

- Given a collection of data objects
 - Each object is associated with a number of features
 - A measure of **similarity** between objects is defined
- **Goal: Group the objects** into clusters such that
 - Objects within each cluster are similar
 - Objects in separate clusters are less similar



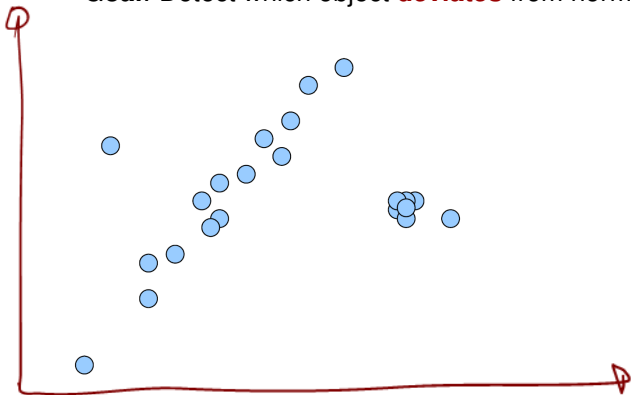
Example: Clustering

Document clustering

- Goal
 - Find groups of similar documents based on the words appearing in them.
- Approach
 - Identify frequently occurring words in each document
 - Define a similarity measure based on the word frequencies
 - Perform clustering to find groups of documents
- Motivation
 - Use the clusters to relate a new document to existing documents
 - Better search algorithms: Return documents that are similar but do not have the exact search keywords

Anomaly detection: Definition

- Given a collection of data objects
 - Each object is associated with a number of features
- **Goal:** Detect which object **deviates** from normal behavior



Example: Anomaly detection

- Credit card **fraud detection**
 - Recognize dubious behavior of credit card transactions based on the transaction history of the card holder.
- Detect **outliers** in data measurement behavior
 - Remove erroneous measurements due to misreading from an instrument
- **Fault detection** in system health monitoring
 - Detect when a wind turbine performs poorly due to ice coating on blades

Representation learning

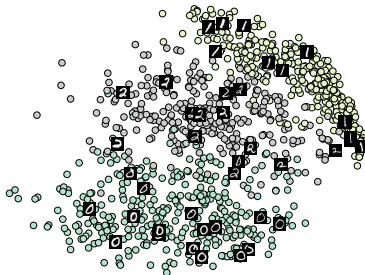
- There are some **latent factors** that can describe the given dataset.
 - Examples: The illumination/hair color/facial expression of human faces, the size/color/orientation of an object, the property X of a molecule, etc.
- **Goal:** Find the latent representations
 - Encode each data point with a *unique* representation
 - This is the foundation of generative AI (image/video generation, LLMs, etc).

Representations

Input data

2
1
0
1
2
1
2
1
2
2

Encoder
 $Enc : \mathbb{R}^D \rightarrow \mathbb{R}^d$

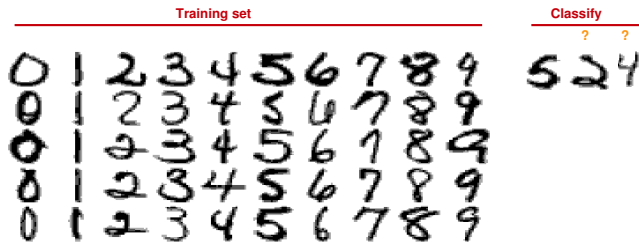


Reconstructions

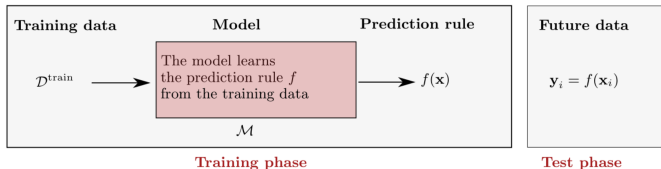
1
1
0
1
1
1
2
1
2
2

Decoder
 $Dec : \mathbb{R}^d \rightarrow \mathbb{R}^D$

Models in machine learning



Classifying a digit is a function: $f : \mathbb{R}^M \rightarrow \{0, 1, \dots, 9\}$

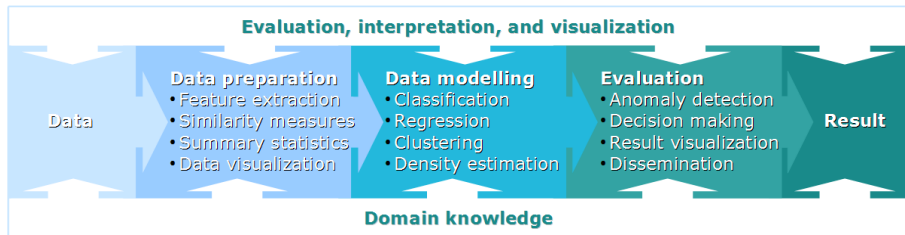


How often the function f makes a mistake on the future/unseen data is called the **generalization error**.

Break

14:45

What is data? [Chapter 2]



- Garbage IN, garbage OUT!

What is data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Also known as variable, features, characteristic, or features.
- Collection of attributes describe an object
 - Also known as a record, point, case, sample, entity, or instance.

Data objects

Attributes

ID	Age	Gender	Name
1	31	F	Alex
2	24	M	Ben
3	52	F	Cindy
4	35	M	Dan
5	58	M	Eric
6	46	F	Fay
7	42	M	George

Discrete / continuous attributes

- **Discrete**

- Finite (or countable infinite) set of values
 - Zip codes
 - Counts
 - Set of words in a collection of documents
- Often represented as integer values

- **Continuous**

- Has real numbers as attribute values
- Examples
 - Temperature
 - Height
 - Weight
- Often represented as floating point variables

Types of attributes

- **Nominal:** Objects belong to a category (equal / not equal)
 - ID numbers
 - Eye color
 - Zip codes
- **Ordinal:** Object can be ranked (greater than / less than)
 - Taste of potato chips on a scale from 1-10
 - Grades
 - Height in {short, medium, tall}
- **Interval:** Distance between object can be measured (addition / subtraction)
 - Calendar dates
 - Temperature in Fahrenheit and Celcius
- **Ratio:** Zero means absence of what is measured (multiplication / division)
 - Length
 - Time
 - Counts
 - Temperature in Kelvin

Qualitative

Quantitative



Discussion

Classify the following attributes

- a) Military rank
- b) Angles measured in degrees
- c) A person's year of birth
- d) A person's age in years
- e) Coat check number
- f) Distance from center of campus
- g) Number of patients in a hospital

- **Discrete**
 - Finite (or countably infinite) set of values
 - **Continuous**
 - Real number
-
- **Nominal** (Equal / Not equal)
 - Objects belong to a category
 - **Ordinal** (Greater than / Less than)
 - Objects can be ranked
 - **Interval** (Addition / Subtraction)
 - Distance between objects can be measured
 - **Ratio** (Multiplication / Division)
 - Zero means absence of what is measured



Discussion

Classify the following attributes

- a) Military rank
Discrete | Ordinal
- b) Angles measured in degrees
Continuous | Ratio*
- c) A person's year of birth
Discrete | Interval
- d) A person's age in years
Discrete | Ratio
- e) Coat check number
Discrete | Nominal (your perspective)
From restaurant's perspective perhaps
Discrete | Interval
- f) Distance from center of DTU campus
Continuous | Ratio
- g) Number of patients in a hospital
Discrete | Ratio

- **Discrete**
 - Finite (or countably infinite) set of values
 - **Continuous**
 - Real number
-
- **Nominal** (Equal / Not equal)
 - Objects belong to a category
 - **Ordinal** (Greater than / Less than)
 - Objects can be ranked
 - **Interval** (Addition / Subtraction)
 - Distance between objects can be measured
 - **Ratio** (Multiplication / Division)
 - Zero means absence of what is measured

Quiz 03: Attribute types

No.	Attribute description	Abbrev.
x_1	Type (0 = served cold, 1 = served hot)	TYPE
x_2	Calories per serving	CAL
x_3	Grams of protein	PROT
x_4	Grams of fat	FAT
x_5	Milligrams of sodium	SOD
x_6	Grams of dietary fiber	FIB
x_7	Grams of complex carbohydrates	CARB
x_8	Grams of sugars	SUG
x_9	Milligrams of potassium	POT
x_{10}	Vitamins and minerals in 0%, 25%, or 100% of FDA recommendations	VIT
x_{11}	Shelf position (1, 2, or 3, counting from the floor)	SHELF
x_{12}	Weight in ounces of one serving	WEIGHT
x_{13}	Number of cups in one serving	CUPS
x_{14}	Name of cereal brand	NAME
y	Average rating of the cereal (from 0 to 100)	RAT

Table 1: Attributes in a study of cereals (i.e. breakfast products, taken from <http://lib.stat.cmu.edu/DASL/Datafiles/Cereals.html>).

In a study of healthy breakfast habits 77 cereal brands were investigated. The attributes of the data are given in Table 1. There are a total of 14 attributes denoted x_1 – x_{14} and one output variable y which defines the average rating of the cereal products by the consumers.

Which statement about the attributes in the data set is incorrect?

- A. NAME is discrete and nominal.
- B. PROT, FAT and SOD are all continuous and ratio.
- C. TYPE and VIT are both discrete and ordinal.
- D. An attribute that is ratio will also be interval.
- E. Don't know.

No.	Attribute description	Abbrev.
x_1	Type (0 = served cold, 1 = served hot)	TYPE
x_2	Calories per serving	CAL
x_3	Grams of protein	PROT
x_4	Grams of fat	FAT
x_5	Milligrams of sodium	SOD
x_6	Grams of dietary fiber	FIB
x_7	Grams of complex carbohydrates	CARB
x_8	Grams of sugars	SUG
x_9	Milligrams of potassium	POT
x_{10}	Vitamins and minerals in 0%, 25%, or 100% of FDA recommendations	VIT
x_{11}	Shelf position (1, 2, or 3, counting from the floor)	SHELF
x_{12}	Weight in ounces of one serving	WEIGHT
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- A. NAME is discrete and nominal. ✓
- B. PROT, FAT and SOD are all continuous and ratio. ✓
- ☒ C. TYPE and VIT are both discrete and ordinal. ✗
- D. An attribute that is ratio will also be interval. ✓
- E. Don't know.

Discrete

- Finite (or countably infinite) set of values

Continuous

- Real number

Nominal (Equal / Not equal)

- Objects belong to a category

Ordinal (Greater than / Less than)

- Objects can be ranked

Interval (Addition / Subtraction)

- Distance between objects can be measured

Ratio (Multiplication / Division)

- Zero means absence of what is measured

Solution:

There are a finite set of brands thus NAME is discrete and as the only operators that can be applied to NAME is equal or not equal NAME is nominal. PROT, FAT and SOD are all continuous and since they have that zero means absence they are ratio. TYPE and VIT are both discrete, however, TYPE is not ordinal, i.e. Hot is not better than Cold, thus

TYPE must be considered nominal, VIT on the other hand is ordinal as 0% is less than 25% which in turn is less than 100%. An attribute that is ratio will also be both interval, ordinal and nominal, i.e. we can apply all the operations $=, \neq, >, <, +, -, *, /$ to a ratio attribute.

Types of data sets

- **Record / tabular**
 - Collection of data objects and their attributes
 - Representation: Table
- **Relational data**
 - Collection of data objects and their relation
 - Representation: Graph
- **Ordered data**
 - Ordered collection of data objects
 - Representation: Sequence

Example: Record example—market basket data

- **Transaction data table**

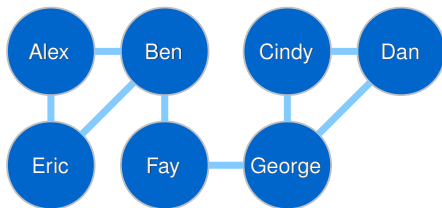
ID	Items
1	Bread, Soda, Milk
2	Beer, Bread
3	Beer, Soda, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Soda, Diaper, Milk

- **Matrix**

ID	Bread	Soda	Milk	Beer	Diaper
1	1	1	1	0	0
2	1	0	0	1	0
3	0	1	1	1	1
4	1	0	1	1	1
5	0	1	1	0	1

Example: Relational data— who knows who?

• Graph

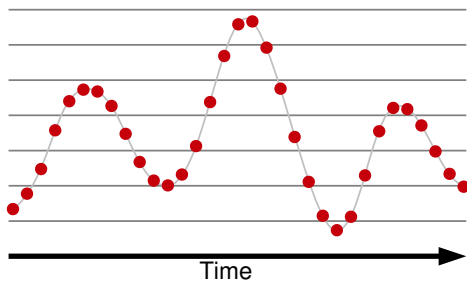


• Matrix

	A	B	C	D	E	F	G
A	0	1	0	0	1	0	0
B	1	0	0	0	1	1	0
C	0	0	0	1	0	0	1
D	0	0	1	0	0	0	1
E	1	1	0	0	0	0	0
F	0	1	0	0	0	0	1
G	0	0	1	1	0	1	0

Example: Sequential data / time series

• Sequence



• Matrix

Time	Value
0	1.3
1	1.8
2	2.5
3	3.6
4	4.4
5	4.7
6	4.6
7	4.3
8	2.4
9	2.1
10	2.0
11	2.3
12	3.1

Quality

- **Data is of high quality if they**
 - Are fit for their intended purpose
 - Correctly represent the phenomena they correspond to
- **Examples of quality problems**
 - Noise
 - Outliers
 - Missing values



Noise

- **Definition**

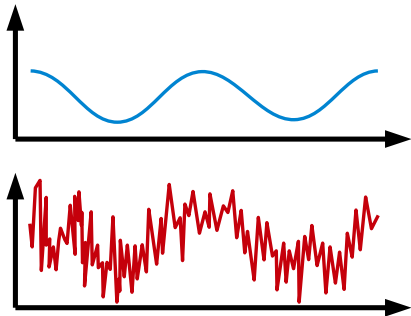
- Unwanted perturbation to a signal
- Unwanted data

- **Causes of noise**

- Limits in measurement accuracy
- Interference from other signals
- Measurement of attributes not related to the data modelling task

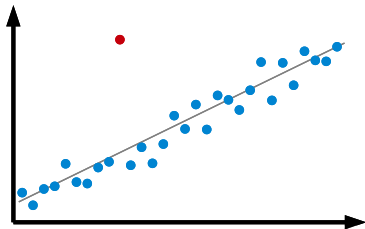
- **Handling noise**

- Exclude noisy attributes
- Remove noise by filtering
- Model the noise (with explicit assumptions)



Outliers

- **Definition**
 - Data object which are significantly different from most others
 - Unwanted data
- **Causes of outliers**
 - Measurement error
 - Natural property of the data
- **Handling outliers**
 - Identify and exclude outliers
 - Model the outliers



Missing values

- **Definition**

- No value is stored for an attribute in a data object (or a specific value to indicate missing values)

- **Causes of missing values**

- Information not collected or measures (e.g. people decide to give their age or gender)
- Attribute is not applicable (e.g. annual income is not relevant for a child)

- **Handling missing values**

- Remove data objects
- Eliminate attributes
- Estimate missing values (e.g. an average)
- Ignore the missing value in analysis
- Model the missing value

ID	Age	Gender	Name
1	31	F	Alex
2	(?)	M	Ben
3	52	F	Cindy
4	35	(?)	Dan
5	(?)	M	Eric
6	(?)	F	Fay
7	42	M	(?)



Discussion

- A group of people were asked to write how many children they have
- Their response was this

3 1 NONE 2 7 3 15 0 1 3 2 zero ~~7~~ 0 1

- A reserach assistant typed the results into a table - His table looked like this

Children	3	1	0	2	7	5	15	0	1	3	-2	0	0	0	1
----------	---	---	---	---	---	---	----	---	---	---	----	---	---	---	---

- Are there any data quality issues?
 - Noise?
 - Outliers?
 - Missing values
- Why have these issues occurred, and how should they be handled?

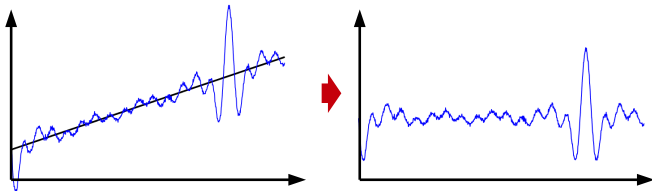
Dataset manipulations

- **Sampling**
 - Selecting a representative subset of the data
- **Feature subset selection**
 - Choose a subset of attributes
- **Feature extraction / transformation**
 - Create new features from existing features
 - Discretization and binarization
 - Apply a fixed transformation to an attribute
 - Aggregation of several attributes into a single attribute
- **Dimensionality reduction**
 - Project data to a low-dimensional subspace

Feature processing

- Eliminating, suppressing, or attenuating certain aspects of the data
 - Noise removal in audio signals
 - Elimination of common words in text documents
 - Removal of background in images
 - Removal of examples which are corrupted
 - De-trending data (if it is not stationary)

Example of de-trending data



Common feature transformation

ID	MPG	Cylinders	Horsepower	Weight	Year	Safety	Acceleration	Origin
1	18	8	150	3436	70	4	11	France
2	28	4	79	2625	82	4	18.6	USA
3	26	4	79	2255	76	3	17.7	USA
3	29	4	70	1937	76	1	14.2	Germany
4	NaN	8	175	3850	70	2	11	USA
5	24	4	90	2430	70	3	14.5	Germany
6	17.5	6	95	3193	76	4	17.8	USA
7	25	4	87	2672	70	-100	17.5	France
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
142	15	8	198	4341	70	2	10	USA

$$\mathbf{X} = \begin{bmatrix} 18 & 8 & 150 & 3436 & 70 & 4 & 11 & 3 \\ 28 & 4 & 79 & 2625 & 82 & 4 & 18.6 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 15 & 8 & 198 & 4341 & 70 & 2 & 10 & 1 \end{bmatrix}$$

Standardize:

$$\mathbf{X} = \begin{bmatrix} \cdots & (X_{1j} - \hat{\mu}_j)/\hat{\sigma}_j & \cdots \\ \cdots & (X_{2j} - \hat{\mu}_j)/\hat{\sigma}_j & \cdots \\ & \vdots & \\ \cdots & (X_{Nj} - \hat{\mu}_j)/\hat{\sigma}_j & \cdots \end{bmatrix}$$

$$\hat{\mu}_j = \frac{1}{N} \sum_{i=1}^N X_{ij}, \quad \hat{\sigma}_j = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_{ij} - \mu_j)^2}$$

Binarize/threshold:

$$\mathbf{X} = \begin{bmatrix} \cdots & 1_{[\theta, \infty[}(x_{1j}) & \cdots \\ \cdots & 1_{[\theta, \infty[}(x_{2j}) & \cdots \\ & \vdots & \\ \cdots & 1_{[\theta, \infty[}(x_{Nj}) & \cdots \end{bmatrix}$$

$$1_{[\theta, \infty[}(x) = 1 \text{ if } x \geq \theta \text{ otherwise } 0$$

Example: One-out-of- K encoding

One-out-of-K coding

	Age	Height	Weight	Nationality
$X =$	-0.2248	-0.4762	-0.2097	'Sweden'
	-0.5890	0.8620	0.6252	'Sweden'
	-0.2938	-1.3617	0.1832	'Sweden'
	-0.8479	0.4550	-1.0298	'Sweden'
	-1.1201	-0.8487	0.9492	'Norway'
	2.5260	-0.3349	0.3071	'Norway'
	1.6555	0.5528	0.1352	'Norway'
	0.3075	1.0391	0.5152	'Norway'
	-1.2571	-1.1176	0.2614	'Norway'
	-0.8655	1.2607	-0.9415	'Sweden'
	-0.1765	0.6601	-0.1623	'Norway'
	0.7914	-0.0679	-0.1461	'Denmark'
	-1.3320	-0.1952	-0.5320	'Denmark'
	-2.3299	-0.2176	1.6821	'Sweden'
	-1.4491	-0.3031	-0.8757	'Sweden'
	0.3335	0.0230	-0.4838	'Sweden'
	0.3914	0.0513	-0.7120	'Denmark'
	0.4517	0.8261	-1.1742	'Sweden'
	-0.1303	1.5270	-0.1922	'Norway'
	0.1837	0.4669	-0.2741	'Denmark'



Age	Height	Weight			
-0.2248	-0.4762	-0.2097	0	0	1
-0.5890	0.8620	0.6252	0	0	1
-0.2938	-1.3617	0.1832	0	0	1
-0.8479	0.4550	-1.0298	0	0	1
-1.1201	-0.8487	0.9492	0	1	0
2.5260	-0.3349	0.3071	0	1	0
1.6555	0.5528	0.1352	0	1	0
0.3075	1.0391	0.5152	0	1	0
-1.2571	-1.1176	0.2614	0	1	0
-0.8655	1.2607	-0.9415	0	0	1
-0.1765	0.6601	-0.1623	0	1	0
0.7914	-0.0679	-0.1461	1	0	0
-1.3320	-0.1952	-0.5320	1	0	0
-2.3299	-0.2176	1.6821	0	0	1
-1.4491	-0.3031	-0.8757	0	0	1
0.3335	0.0230	-0.4838	0	0	1
0.3914	0.0513	-0.7120	1	0	0
0.4517	0.8261	-1.1742	0	0	1
-0.1303	1.5270	-0.1922	0	1	0
0.1837	0.4669	-0.2741	1	0	0

Example: Bag-of-words representation

- First three sentences on wikipedia.org
 - The bag-of-words model is a simplifying assumption used in natural language processing and information retrieval
 - In this model, a text (such as a sentence or a document) is represented as an unordered collection of words, disregarding grammar and even word order
 - The bag-of-words model is used in some methods of document classification



Example: Bag-of-words representation

- The bag-of-words model is a simplifying assumption used in natural language processing and information retrieval
- In this model, a text (such as a sentence or a document) is represented as an unordered collection of words, disregarding grammar and even word order
- The bag-of-words model is used in some methods of document classification



We will treat **this text** as a data set and create a bag-of-words model of it



Example: Bag-of-words representation

- Elimination of common words (so-called stop words)
 - The bag-of-words model is a simplifying assumption used in natural language processing and information retrieval
 - In this model, a text (such as a sentence or a document) is represented as an unordered collection of words, disregarding grammar and even word order
 - The bag-of-words model is used in some methods of document classification

bag-of-words model
methods document
classification



Example: Bag-of-words

- Representation as a matrix

Word	Sentence		
	1	2	3
bag-of-words	1		1
model	1	1	1
simplifying	1		
assumption	1		
natural	1		
language	1		
processing	1		
information	1		
retrieval	1		
text		1	
sentence		1	
document		1	1
represented		1	
unordered		1	
collection		1	
words		1	
disregarding		1	
grammar		1	
word		1	
order		1	
methods			1
classification			1

Example: Bag-of-words

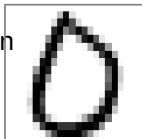
- Stemming

Word	Sentence		
	1	2	3
bag-of-word*	1		1
model*	1	1	1
simplif*	1		
assum*	1		
natural*	1		
languag*	1		
process*	1		
information*	1		
retriev*	1		
text*		1	
sentence*		1	
document*		1	1
represent*		1	
unorder*		1	
collect*		1	
word*		2	
disregard*		1	
grammar*		1	
order*		1	
method*			1
classif*			1

Example: Image representation

- **Example: Handwritten digits**

- Preprocessing
 - Digitalization
 - Centering
 - Rotation
 - Scaling



28×28

$$M_0 = \begin{bmatrix} 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & \dots & 0.3 & 1 & 0.2 & 0 & \dots & 0 \\ \vdots & & & & & & & \vdots \\ 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$



- Vectorization 1×784 $x_0 = [0 \dots 0 \ 0.3 \ 1 \ 0.2 \ 0 \dots 0]^T$

- Matrix representation of data set

$$X = \begin{bmatrix} \text{---} x_1 \text{---} \\ \text{---} x_2 \text{---} \\ \vdots \\ \text{---} x_N \text{---} \end{bmatrix}$$





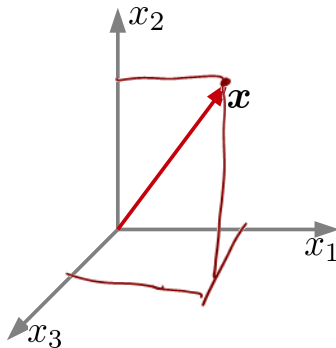





If each image is 28×28 pixels
then X is a $N \times 784$ matrix.

Key concept: Data in a vector space

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix}$$



Visualization

Visualization

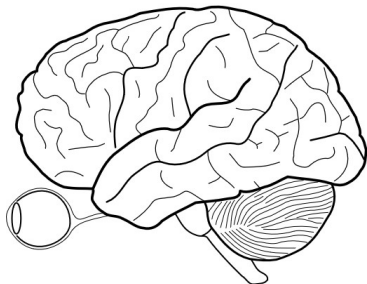
The drawing shows me at one glance what might be spread over ten pages in a book."

- Ivan S. Turgenev's novel Fathers and Sons, 1862.

Use a picture. It's worth a thousand words."

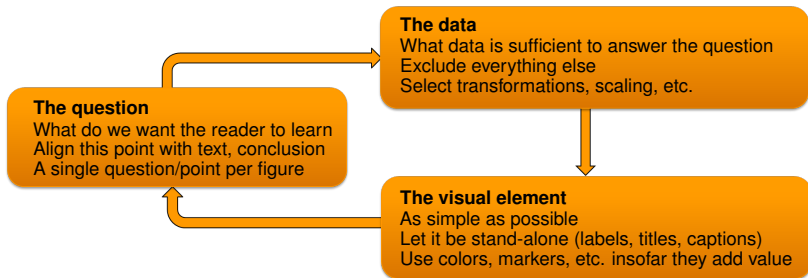
- Arthur Brisbane to the Syracuse Advertising Men's Club, in March 1911

- A main function of the brain is to process visual information
- We can exploit this capacity using visualization of the data in order to:
 - Detect new patterns, i.e. exploratory data analysis)
 - **Dissiminate results, i.e. visualizations/plots in written work (today)**
- We should take into account how the brains visual system works



Illustrations as technical writing

- The purpose of the text is to communicate an idea (**vs. plots has a purpose**)
- Be grammatically correct (**vs. elementary “rules” of good plotting**)
- Ensure the text is readable (**vs. labels, legends or lines nobody can read**)
- Avoid long/complicated paragraph (**vs. plots that are overly complicated**)
- Don't lie or exaggerate (**vs. distort data in a plot**)



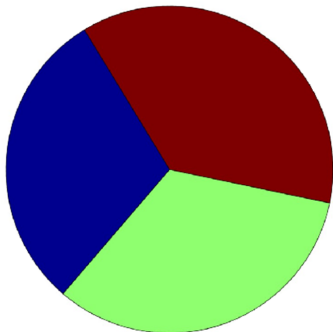
Important choices for visualizations

- **Representation:** How will you map objects, attributes, and relations to visual elements?
 - Positions, lengths, colors, areas, orientation
- **Arrangement** How will you display the visual elements?
 - Viewpoint, transparency, separation, grouping
- **Selection:** How will you handle a large number of attributes and data objects?
 - Display a subset, focus on a region of interest, show summaries

Representation

Area represents proportion

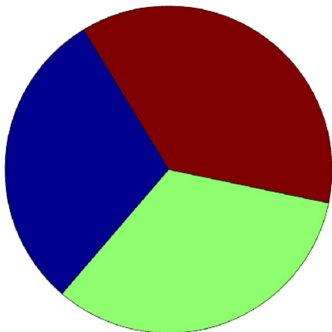
- Which is smallest, middle, and largest?
- What are the proportions approximately?



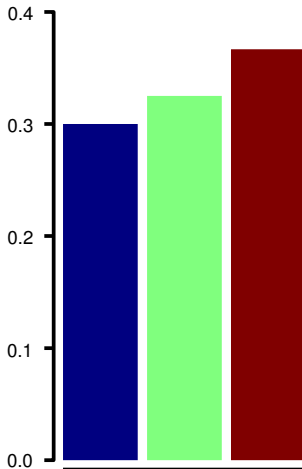
Representation

Area represents proportion

- Which is smallest, middle, and largest?
- What are the proportions approximately?



- Height represents proportion



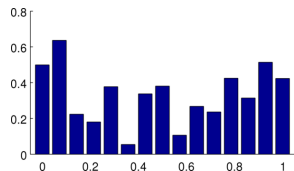
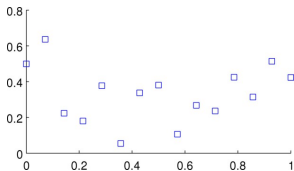
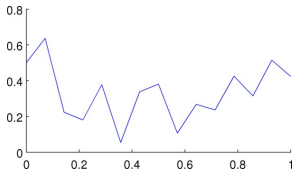
Selection

- Elimination or de-emphasis of certain objects or attributes
- A subset of **attributes**
 - **Why?** A graph can only show so many attributes – focus on the relevant
 - How?
 - Dimensionality reduction
 - Plot pairs of attributes
- A subset of **objects**
 - **Why?** A graph can only show so many objects – focus on the relevant
 - Random sampling
 - Display of region of interest
 - Use density estimation

Types of plots

- **Distribution of a single attribute**
 - Histogram
 - Empirical cumulative distribution
 - Percentile plots
 - Box plot
- **Relation among attributes**
 - 2D histogram
 - Heat maps and contour plots
 - Scatter plots
- **Visualization of high-dimensional objects**
 - Matrix plots
 - Parallel coordinates
 - Star plots

Basic plots



The Iris Dataset

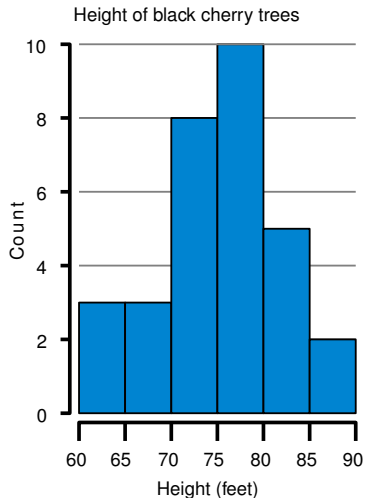
- **Three types of Iris flowers**
 - 50 instances of each class, 150 in total
- **Attributes**
 - Sepal (outermost leaves)
 - length in cm
 - width in cm
 - Petal (innermost leaves)
 - length in cm
 - width in cm
 - Class of flower
 - Iris Setosa
 - Iris Versicolour
 - Iris Virginica

Flower ID	Attribute			
	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
.	.	.	.	.
.	.	.	.	.
150	5.9	3.0	5.1	1.8

Distribution of a single attribute

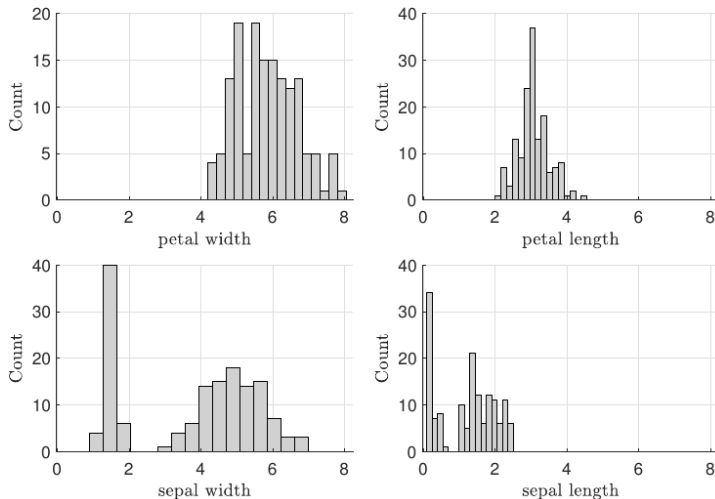
Histograms

- Shows distribution of a single variable
 - Divide the values into bins
 - Bar plot of the number of values in bin
 - Height indicates count of values
 - Shape determined by
 - Distribution of data
 - Number of bins / bin width

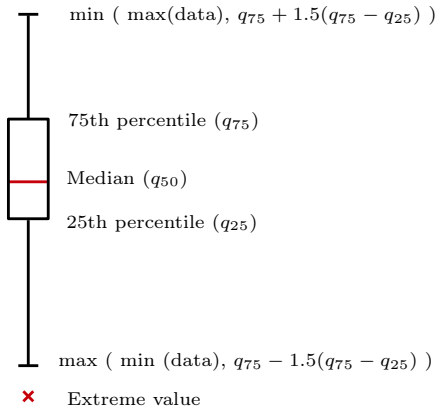


$\mathbf{x}^T = [60, 64, 64, 66, 67, 69, 71, 72, 72, 72, 72, 73, 74, 74, 75, 75, 76, 76, 76, 77, 77, 78, 78, 79, 80, 80, 81, 82, 84, 85, 85, 89]$

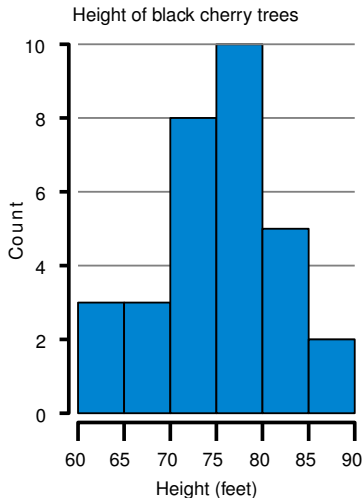
Histograms of the Iris data attributes



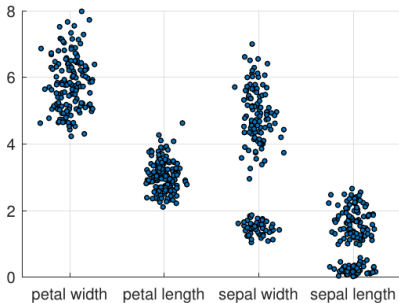
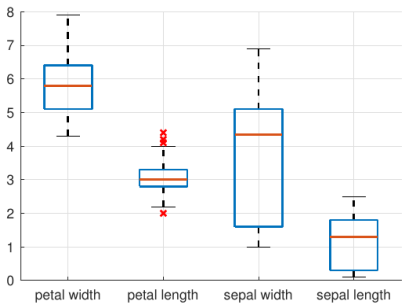
Boxplots



The plotted whisker extends to the adjacent value, which is the most extreme data value that is not an outlier.



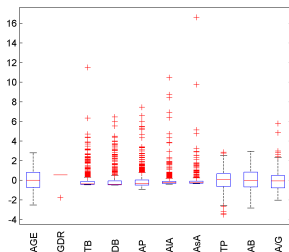
Boxplots



Quiz 04: Boxplots

No.	Attribute description	Abbrev.
x_1	Age (in years)	AGE
x_2	Gender (Female=0, Male=1)	GDR
x_3	Total Bilirubin	TB
x_4	Direct Bilirubin	DB
x_5	Alkaline Phosphotase	AP
x_6	Alamine Aminotransferase	AIA
x_7	Aspartate Aminotransferase	AsA
x_8	Total Protiens	TP
x_9	Albumin	AB
x_{10}	Albumin to Globulin ratio	A/G
y	0=No liver disease, 1=Liver disease	LD

Table 1: Liver disease dataset.



The attributes x_1 – x_{10} are standardized (i.e., the mean has been subtracted each attribute and the attributes divided by their standard deviations). The figure shows a boxplot for the standardized data. Which of the following statements is *correct*?

- A. The value of the 50th and 75th percentiles of the attribute DB coincides.
- B. Even though the distribution of AIA and AsA may have a similar shape this does not imply that the two attributes are correlated.
- C. The attribute TB is likely to be normal distributed.
- D. The attribute GDR has a clear outlier that should be removed.
- E. Don't know.

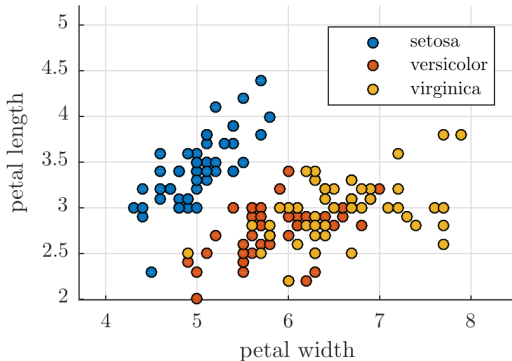
The 25th and 50th percentile but not the 50th and 75th percentiles of the attribute DB coincides. A1A and AsA will not necessarily be highly correlated even though their distributions may have a similar shape (hence, this is correct). For attributes to be correlated it is important they take on high or low values systematically, however, this can not be inspected in

a boxplot. TB is not likely to be normal distribution as this attribute does not have a symmetric but highly right skewed distribution. The attribute GDR does not have a clear outlier, in fact the outlier corresponds to the females in the dataset and all we can deduce from the plot is that more than 75 % of the observations are males.

Relation between attributes

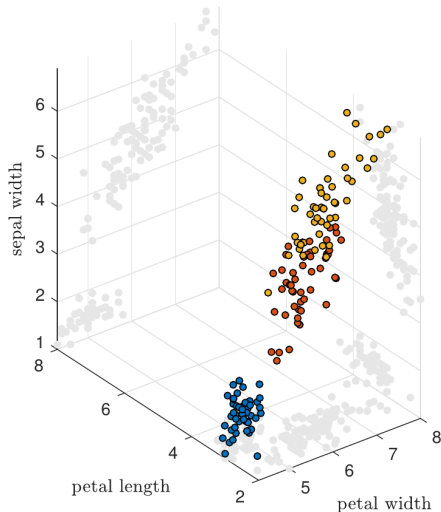
Scatter plots

- Shows **relations** between attributes
 - Assess dependence between attributes
 - Used with classes to assess separability



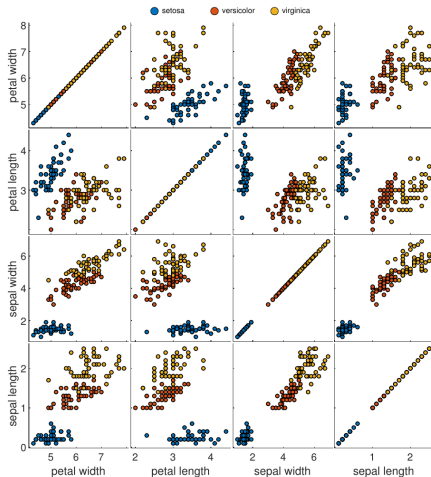
Scatter plots

- Shows **relations** between attributes
 - Assess dependence between attributes
 - Used with classes to assess separability
 - 3D plots are often confusing; avoid if possible



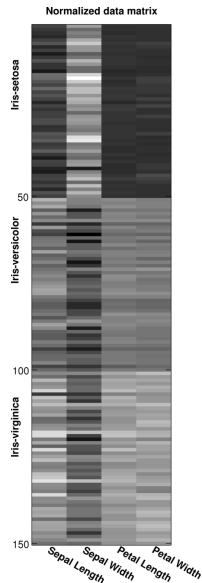
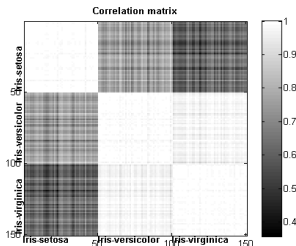
Scatter plots

- Scatter plot matrix — all pairs of attributes



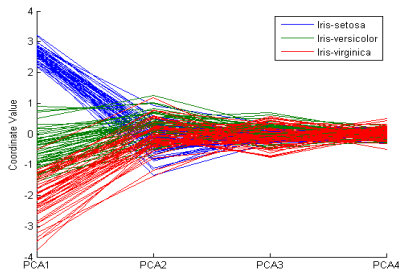
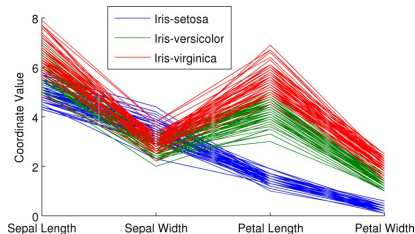
Matrix plots

- Plot of raw data matrix
 - Useful when objects are sorted according to class
 - Typically, attributes are normalized
- Plots of similarity matrices
 - Useful for visualizing the relation between objects



Parallel coordinates

- **Plot high-dimensional data**
- Instead of perpendicular axes
 - Use parallel axes
- Attribute values are plotted as a point
 - and the points are connected by a line
- Each object is represented as a line
- Lines representing a group of objects
 - Are similar in some sense
 - Ordering of attributes is important in seeing such groupings



ACCENT

- **Apprehension**
 - Is it easy to see what is important in the graph?
- **Clarity**
 - Are the most important elements visually most prominent?
- **Consistency**
 - Have you used the same colors, shapes, etc. as in other graphs?
- **Efficiency**
 - Does it convey its information in the most simple and efficient way?
- **Necessity**
 - Are all elements of the graph necessary to represent data?
- **Truthfulness**
 - Does the graph represent the data correctly?

Tufte's guidelines

Graphical excellence

- Well-designed presentation of interesting data – a matter of
 - substance, statistics, and design
- Complex ideas communicated with
 - clarity, precision, and efficiency
- Gives the viewer
 - the greatest number of ideas
 - in the shortest time
 - with the least ink
 - in the smallest place.
- Nearly always multivariate
- Requires telling the truth about the data
- Maximise Data-ink ratio:

$$\text{Data-ink ratio} = \frac{\text{Data-ink}}{\text{Total ink used}}$$



Summary

Exercises and Assignments

- **From theory to practice:** Learn concepts, algorithmic insights, intuition through experimentation, practical ML skills and strengthen problem-solving and coding skills.
- Recommendations regarding **generative AI** for exercises and assignments.
- **Assignments (mandatory)** are included in the weekly exercise notebooks (8 out of 12 must be approved/passed)
 - **Note:** Assignments can not always be completed in the 2-hour timeslot - that is intentional!

(Demo if time permits)

Exercise rooms

A082

- **B116-A081** [General engineering, Elektrotek, **all**] [137]
- **B116-A083** [Cybertek, Infotek, Produktion og Konstruktion, Design og Innovation, **all**] [50]
- **B116-H012** [MatTek, MMC **only**] [54]
- **B116-H013** [Softwaretek **only**] [30]
- **B116-H015** [Medicin og Tek **only**] [30]
- **B116-H016** [Data science og mgn, Business analytics, **all**] [50]
- **B116-H019** [Bioinfo og systembio, Bio tek, Bæredygtigt energidesign, Miljøtek, Life Science, Chemistry, **all**] [50]
- **B116-H040** [KID **only**] [50]
- **B116-H042** [KID **only**] [40]
- **B116 South Lobby** [**all**, B116-A081 overflow] [123]
- **B116 East Lobby** [**all**] [66]
- **B116 North Lobby** [**all**] [48]

See map at <https://www.timeedit.net/dtu>

